Asher Straus

624 Ontario Ave, Tampa, FL 33606 | asherstraus2027@u.northwestern.edu | (914) 924-6220
Portfolio: **strausasher.github.io/Portfolio** | LinkedIn: **linkedin.com/in/asher-straus-0330452a9**

EDUCATION

Northwestern University | McCormick School of Engineering | Evanston, IL — Sept 2023 – Mar 2028 (expected)
B.S. Manufacturing & Design Engineering | *Minor in Art* | *Segal Design Certificate* | *Robotics Certificate*
GPA: 3.6 / 4.0 cumulative · 3.9 / 4.0 since sophomore year

WORK EXPERIENCE

Siemens Healthineers — *Mechanical Engineering Co-op, Molecular Imaging · Hoffman Estates, IL* — Jun 2026 – Present
- Updated CAD models and drawings for mechanical components in SPECT and PET scanners to resolve interferences and improve manufacturability. Applied GD&T to fully define part requirements.
- Designed fixtures and brackets in Siemens NX/Designcenter. Used 3D printing to evaluate and iterate on design.
- Contributed to the development of a multi-axis stage and investigated motorized drives to support SPECT research team.

P1 Motor Club — *Virtual Test Driver, Research & Promotions Intern · Tampa, FL* — Apr 2023 – Present
- Test new track layouts in simulators and give performance feedback to engineers to inform design changes.
- Promote the club and its track property to prospective members, including by building a large-scale, mostly 3D-printed model from the civil engineer's Civil 3D files, sized to travel in a truck as a membership sales tool.

MIT Beaverworks Summer Institute — *Teaching Assistant, "Basics of ASICs" · Cambridge, MA* — Jun 2024 – Aug 2024
- Guided students through ASIC schematic design, simulation, and testing, and helped refine labs and curriculum.

Monticello Motor Club — *Go-Kart Instructor, Technician & Social Media Coordinator · Monticello, NY* — Jun 2018 – Jul 2022
- Diagnosed and repaired kart systems for safe, reliable operation; coached drivers of all skill levels.

RESEARCH & PROJECTS

Advanced Intelligent Manufacturing Lab — *Researcher · NSF ERC-HAMMER · MSEC 2026 Poster Co-author* — Nov 2025 – Apr 2026
- Smart Sheet Smith: co-author on the MSEC 2026 / NAMRC54 poster for a five-agent AI pipeline that turns a 2D sheet-metal drawing into a verified bending process plan, with no 3D CAD or labeled data.
- Built the Tracer Agent's evaluation corpus (CAD parts, multi-view drawings, hand-derived bend math) and a SolidWorks VBA pipeline that generates hundreds of valid drawings, cutting data creation from days to minutes.
- Fabricated 10+ flexible resistive tactile sensors (carbon nanotubes in Ecoflex) and ran near-field electrospinning at 10 kV (Fall 2025).

Formula SAE Brake System — *Suspension Member · Northwestern Formula Racing* — Sep 2024 – Present
- Hand-calculated the pedal force for a 1 G stop (~42 lbf pedal to ~690 lbf tire force), improved the team MATLAB brake sim, and verified rotor limits with ANSYS FEA.
- Machined and lathe-finished rotors, routed lines, and bled the system on the car.

CT Scanner for Aquatic Animals — *Design & Build Engineer · Major Aquarium · Chicago, IL* — Sept 2025 – Mar 2026
- Adapted a medical CT scanner for fish with a veterinarian: stabilization, ventilation and saltwater corrosion protection.

SME Dorm Accessibility Competition Project — *Team Lead · SME National Student Manufacturing Innovation Challenge* — Aug 2026 – Present
- Leading a team of five on a residence-hall accessibility project; in concept stage while we scope budget and ideas.

Midwest EV Openpilot Retrofit — *Enclosure Designer · Pre-Autopilot Tesla Model S retrofit* — 2026 – Present
- Design the enclosure for the custom PCB in a Tesla Model S openpilot retrofit, iterating as the board goes through design review.

Bionic Wrench Assembly Fixture — *Designer & Manufacturing Engineer · DSGN 386* — Mar 2025 – Jun 2025
- Iterated 7 fixture versions; wrote the SOP, value stream map and line balance for a 1M-unit/year plan, with a pilot cycle time of 3:58.

ExtendIt Desk Extension and Assistive Devices — *Designer & Engineer · Project Manager (Misericordia)* — Jan 2024 – Mar 2025
- ExtendIt: locking lecture-hall desk extension (+50% surface) after 22 user interviews and 7 mockups. StimSpin and Pillowscape: sensory devices for adults with cerebral palsy, tested with users.

TECHNICAL SKILLS

Design & Fabrication: Siemens NX/Designcenter, SolidWorks, Onshape, AutoCAD/Civil 3D, GD&T, Engineering Drawings, Fixture & Jig Design, 3D Printing, Laser Cutting, Waterjet, Machining (Lathe/CNC/Mill), Welding, Sand/Iron Casting, Sheet Metal, Metrology, Snap-Fit/Compliant Design, Rapid Prototyping, DFM/DFR

Analysis & Software: Ansys, FEA/Topology Opt., Thermal Sim., MATLAB, Python, VBA, SolidWorks API, LLM Pipelines, Embedded Systems/Raspberry Pi, FMEA, Cost Modeling, Lean/VSM, Line Balancing, Excel, Unity, GitHub, Photoshop, Circuit Design, Radio Control

Relevant Courses: Manufacturing Engineering Design (DFM), Computer Integrated Manufacturing, Geometry & Manufacturing, Mechanical Design & Manufacturing, Materials Selection, Mechatronics, Mechanics of Materials, Electronics Design, Human-Centered Product Design, Optimization

Clubs: Formula SAE, Baja SAE, Combat Robotics, SME Student Chapter